SET #1

CONFIDENTIAL

UTM UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION

SEMESTER II 2020/2021

SUBJECT CODE	**: SCSI/SECI 1013**
SUBJECT NAME	**: DISCRETE STRUCTURE**
YEAR/COURSE	**:**
TIME	**:**
DATE	**: 9TH JULY 2021**
VENUE	**: E-LEARNING**

INSTRUCTIONS TO THE STUDENTS:

1. Please answer ALL the questions in the answer sheet form.
2. Fill in your particular in the answer sheet.
3. A candidate who is suspected of cheating in examination is liable to disciplinary action including (but not limited to) suspension or expulsion from the university. All materials and/or devices which are found in violation of any examination rules and regulation will be confiscated.

NAME	
MATRIC NO.	
SECTION	
LECTURER	

(This question paper consists of 6 pages including this page)

QUESTION 1 **20 MARKS**

a) At one factory, there are three machine that produce microchip for computer. Of their production, machine A, B and C respectively produce 2%, 1% and 3% defective microchip. Of the total production of microchip in the factory, machine A produces 35%, machine B produces 25% and machine C produce 40%.

i) If one microchip is selected at random from the total microchip produced in a day, what is the probability that it is defective?

(7 marks)

ii) If the selected microchip is defective, what is probability that it was produced by machine C?

(3 marks)

b) A drone was built with three-computer system. During the operation, if system (S_1) fails, it is bypassed and system (S_2) is used. If system (S_2) fails, it is bypassed and system (S_3) is used. Suppose that the probability of failure of any one system is 0.15, and assume that the failures of these systems are mutually independent events.

i) What is the probability that the drone does not fail?

(6 marks)

ii) If a fourth system (S_4) is added to increase the reliability of the drone, what is the probability that the drone does not fail?

(4 marks)

QUESTION 2 **20 MARKS**

a) Upon completing all the expected daily Covid-19 vaccine doses, the doctors in charge will analyze all the collected data from 5 stations (denoted as Station 1, Station 2, Station 3, Station 4, Station 5) to ensure all the recorded data and the procedures are sequentially matched. In order to have one distinct record for each recipient, the collected data from these stations are connected to each other. However, there are two distinct data that have been recorded in Station 2, therefore, the doctors need to analyze the data from this station twice before moving to the next data.

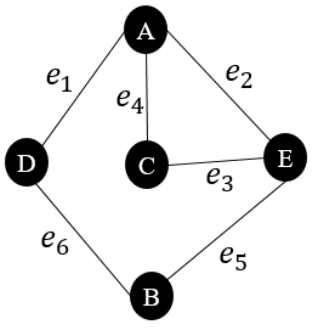
i) Draw an undirected graph representing the data connection. Label all the edges and vertices.

(4 marks)

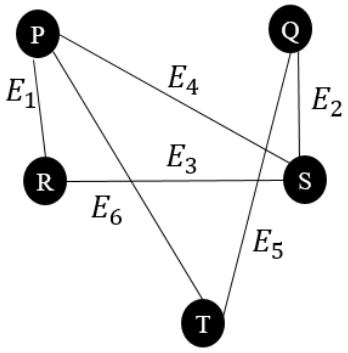
ii) Justify whether the graph in (i) is a regular graph or not.

(2 marks)

b) For the following pair of simple graphs **G** and **H**, determine whether **G** and **H** are isomorphic. If yes, show the adjacency matrix as proof.



Graph **G**



Graph **H**

(7 marks)

c) Identify whether the graph below (**Figure 1**) has Euler circuit or Euler trail. If it does have a Euler circuit or Euler trail, describe one.

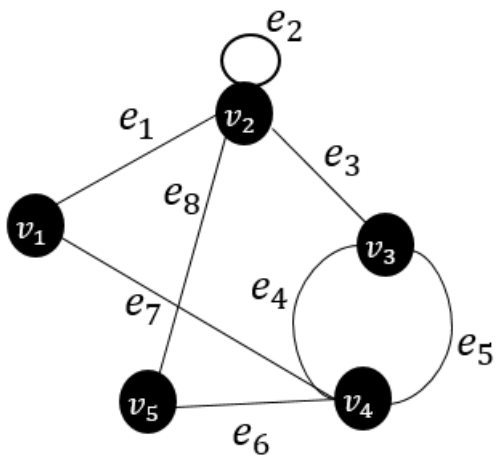


Figure 1

(4 marks)

d) Does the Hamilton circuit exist in graph in **Figure 1** above? If yes, exhibit one. If no, justify your answer.

(3 marks)

QUESTION 3 **20 MARKS**

a) Draw a full binary tree with five internal vertices.

(4 marks)

b) Find three spanning trees from the graph in **Figure 2**.

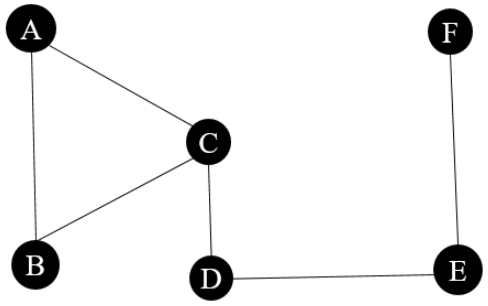


Figure 2

(6 marks)

c) Describe the action of Kruskal's algorithm on the graph shown in **Figure 3** (use **Table 1** as template to show your work).

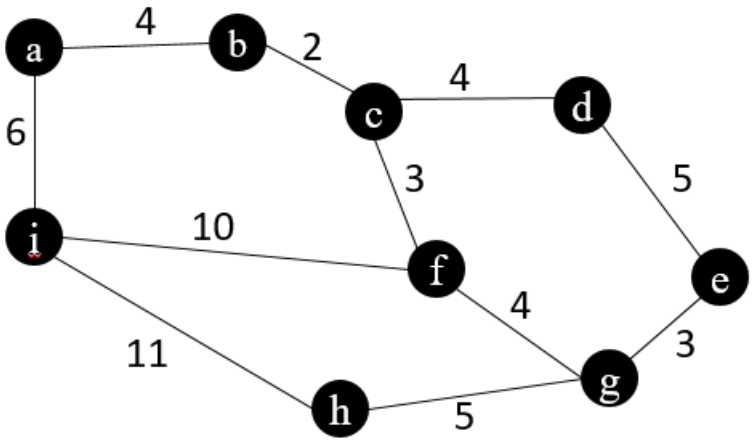


Figure 3

Table 1

Iteration No.	Edge Considered	Weight	Action Taken

(8 marks)

d) Draw the minimum spanning tree produced by Kruskal's algorithm in (c) and state the minimum total weight.

(2 marks)